

A Simple Tutorial on Few-Shot Learning

Terminology at a Glance

- Few-Shot Learning
- Zero-Shot Learning
- One-Shot Learning
- In-Context Learning
- Few-Shot Prompting

What Is Few-Shot Learning?

Few-shot learning means teaching a model to perform a task using only a **small number of examples**, rather than a large labeled dataset.

Example: Sentiment Classification

Example 1:

“This movie was amazing.” → Positive

Example 2:

“The movie was boring.” → Negative

Now classify:

“I really enjoyed this film.” → ?

Expected answer:

Positive

Zero-Shot, One-Shot, and Few-Shot

These three settings differ only in **how many examples** are given to the model before it must answer.

Zero-Shot Example

Instruction: *Classify the sentiment of each review.*

Classify: “*The food was excellent.*”

Note: No prior examples are given — the model must rely only on the instruction itself.

One-Shot Example

Instruction: *Classify the sentiment of each review.*

“The food was terrible.” → Negative

Now: “The food was excellent.” → ?

Note: A single labeled example is given before the new query.

Few-Shot Example

Instruction: *Classify the sentiment of each review.*

“The food was terrible.” → Negative

“The food was okay, nothing special.” → Neutral

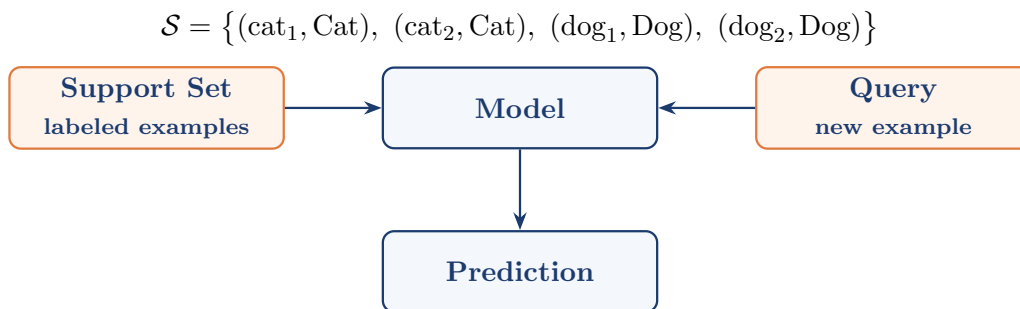
Now: “The food was excellent.” → ?

Note: Several labeled examples are shown before the new query, giving the model a clearer sense of the pattern.

Support Set and Query Set

The **support set** is the small collection of labeled examples used to help the model understand or adapt to the task. The **query** is the new, unlabeled example the model must predict.

Consider an image classification task, where each item in the support set is an *image* paired with its class label:



N -Way K -Shot Learning

Few-shot tasks are often described using the notation:

N -way K -shot

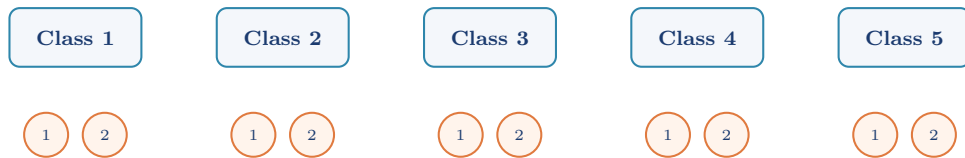
- N = number of classes in the task,
- K = number of support examples given per class.

Example: 5-Way 2-Shot

5-way 2-shot $\implies$ 5 classes, 2 examples per class

The total size of the support set is:

$$N \times K = 5 \times 2 = 10 \text{ examples}$$



In-Context Learning

For modern large language models, few-shot learning is usually done through **in-context learning**: the examples — often called **demonstrations** — are placed directly inside the prompt, and the model is not retrained. When this is done with several demonstrations, it is also called **few-shot prompting**.

```
2 + 3 = 5
4 + 7 = 11
8 + 2 = ?
```

The model predicts 10, simply by recognizing the pattern from the two demonstrations placed in the context.

Prompt = Demonstrations + Query

Crucially, the model's parameters are **not updated**.

Few-Shot Prompting vs. Few-Shot Fine-Tuning

Two different ideas are sometimes both called “**few-shot learning**.” It is important to keep them separate.

Few-Shot Prompting	Few-Shot Fine-Tuning
Examples are placed in the prompt:	Examples are used to actually train the model:
Examples + Query → Prediction	$\theta' = \theta - \eta \nabla_{\theta} L$
↓	↓
No parameter update	Parameter update

Example Selection in Few-Shot Prompting

When many candidate examples are available but only a few can fit in the prompt, we must decide: **which examples should we select?**

Example: Relevance Matters

Task: *Classify the sentiment of each product review.*

Query:

“The battery lasts forever.”

More useful support example (topically related):

“The battery dies after one hour.” → Negative

Less useful support example (unrelated topic):

“The restaurant was terrible.” → Negative

The battery-related example is more useful because it is closer in topic and structure to the query.

Good support examples should ideally be:

- **relevant** to the query,
- **correctly labeled**,
- **diverse**, covering different cases,
- **representative** of the overall task.

The *order* in which examples are presented can also affect an LLM’s predictions. Therefore, a core few-shot learning problem is to **select good examples** and **choose a good order**.

More examples are not always better. A small set of high-quality, relevant examples can outperform a larger set of noisy or poorly chosen ones.

Few-Shot Prompting vs. RAG

Few-shot examples show the model how to perform the task, while **RAG** provides the information needed to perform it.